

Dhruba Jyoti Chakraborty

+91(8906894144) ♦ Bengaluru

Dhrubajyotichakraborty071@gmail.com ♦ linkedin.com/in/dhruvchakraborty ♦ github.com/Dhruv19468

SKILLS

- **Languages:** Python, PySpark, SQL
- **Frameworks:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, SciPy, Statsmodels
- **Data Analysis & Visualization:** Tableau, Power BI
- **Other Skills:** Feature engineering, Hypothesis testing, Statistical Analysis, Machine Learning, Linear/Logistic Regression, Random Forest, Gradient Boosted Decision Tree, Clustering, Dimensionality Reduction, Data Modelling

EXPERIENCE

Senior Engineer (Data scientist)

Bosch Global Software Technologies

Apr 2022 - Present

Bengaluru

Credit Risk Assessment & Credit Score Prediction Model:

- Developed and implemented a credit scoring Machine Learning model using logistic regression to assess the likelihood of customer default.
- Conducted extensive data preprocessing, handling missing values, outlier detection, and ensuring data quality for accurate predictions using SQL, Python and PySpark.
- Engineered and selected features using Information Value (IV) and Gini coefficient to identify variables with maximum predictive power.
- Improved model accuracy through reject inference techniques, addressing bias from previously declined loan applicants.
- Evaluated Machine Learning model performance using ROC curves (AUC>82%), confusion matrices, and Kolmogorov-Smirnov (KS) statistics, achieving high predictive power.
- Ensured model interpretability by analyzing log-odds ratios, providing stakeholders clear insights into factors driving credit risk.
- Retrained model periodically to mitigate drift, generated pickle files, and collaborated with IT for seamless deployment.
- Automated the credit scoring process using Python, enabling faster and more consistent risk assessment across loan applications.
- Collaborated with cross-functional teams, including finance and risk management, to validate model outputs and align with regulations.

Engineer (Data analyst)

Dynamics Technology Limited

Apr 2019 - Apr 2022

Bengaluru

- Performed EDA using Python and Tableau to evaluate data trends, uncover patterns, and support data-driven strategies, while improving problem-solving speed and increasing data processing efficiency by managing expectations and handling large datasets.

Product Profiling for Off-Road Vehicles:

- Led the 'Product Profiling for Off-Road Vehicles' project, preparing, cleaning, and transforming data using Excel, Python and Tableau driving a 25% improvement in team efficiency and 15% more accurate sales projections.
- Provided regular data-driven insights and visualizations to senior leadership, enabling strategic decision-making and optimizing resource allocation across multiple projects.

Data Migration from old ERP to SAP:

- Led the migration of 150,000+ records from an old ERP system to SAP, performing extensive data cleaning, manipulation, and validation using advanced Excel functions (SUM, CONCAT, VLOOKUP), ensuring 100% data accuracy and integrity.
- Collaborated with cross-functional teams to analyze, transform, and standardize legacy data, improving data quality and streamlining the migration process, reducing manual corrections by 30%.

PROJECTS

- **Analyze Walmart's gender-based Black Friday spending behavior**
<https://www.datascienceportfol.io/Dhruv194/projects/8>
 - Preprocessed data using Python and Excel by handling nulls, correcting data types, and visualizing insights.
 - Identified non-overlapping confidence intervals for gender, with males spending more on average, suggesting a focus on male customers while also creating initiatives to attract female shoppers.
 - Found overlapping confidence intervals for age groups (26-35 and 36-45) and marital status, recommending equal emphasis on ads and promotions for these segments to maintain market stability and growth.
- **Analyzing Target Brazil's Order Operations and Customer Insights: 2016-2018**
<https://www.datascienceportfol.io/Dhruv194/projects/10>
 - Completed end-to-end analysis on Target Brazil's e-commerce dataset using Excel and BigQuery, performing EDA to extract insights.
 - Identified a 13,608% increase in order volumes from 2016 to 2017 and a 20% increase in 2018, highlighting rapid e-commerce growth.
 - Discovered a recurring monthly order pattern with peaks in November and declines in December, enabling targeted strategies like discounts or promotions to optimize order volumes year-round.

EDUCATION

- **Scaler**
Specialized in Data Science & Machine Learning
- **Surendra Institute of Engineering & Management**
BE/B.Tech in Mechanical Engineering — 8.3 CGPA